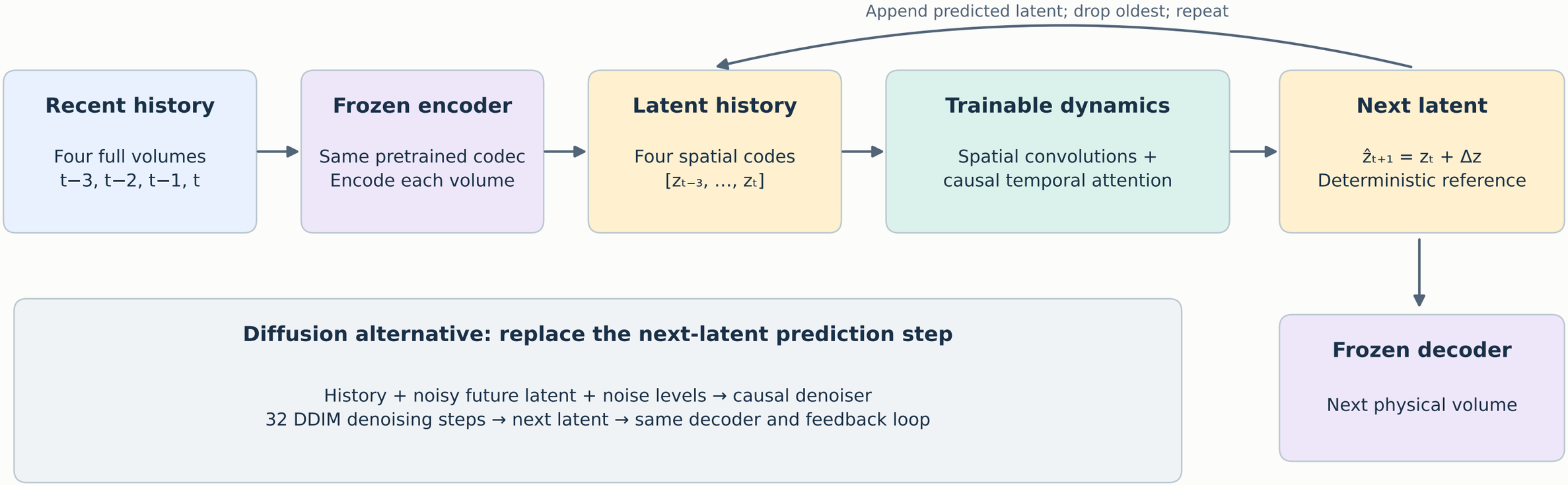
03 / Temporal rollout

Question: can a learned evolution rule advance the compressed plasma state through time?



Deterministic training: four-step rollout loss in decoded physical fields. Encoder/decoder stay frozen.

Diffusion training: denoising loss in normalized latent space; sampled outputs are scored physically.

Latest ablation: next-token loss versus loss across all noised sequence positions.

This diagram shows full-volume forecasting. Sparse-observation transfer is a later conditioning task.

Current sampler predicts one next state at a time; rolling-block Diffusion Forcing is not implemented here.